Stage 1.0 — Range Table Targeting (Powered Ascent + Ballistic Arc)

Era: Redstone/Mercury-Redstone class, ~1961 (fictional program). **Prerequisite:** Phase 0 (all four drills). **Depends on:** 0.0, 0.1 (slide rule, table interpolation) directly; 0.3 (triangle solving) indirectly, for the same "expect more than one valid answer" habit.

This version deliberately simplifies two things — a constant Isp and a given, un-derived loss budget — to keep the first Stage 1 problem to about half an hour. For the harder, more complete version of this same ascent (where both of those are derived rather than assumed), see **1.1** (liftoff sanity check + gravity-turn loss, first-principles) and **1.2** (non-constant Isp, full afternoon).

Narrative frame

You're the flight dynamics engineer for **Project Arrowhead**, a fictional 1961 suborbital test program flying the **Redwing-3** booster out of a fictional coastal range. Range Safety has assigned tomorrow's flight an impact zone centered **210.0 nautical miles** downrange of the pad, inside a larger safety corridor. Your job: using the vehicle's known performance and a burnout altitude fixed by the guidance program, build the range table for this vehicle and tell the pad crew what burnout flight path angle(s) will put the impact where Range Safety wants it — and confirm there's more than one way to get there.

Given data

Vehicle performance (powered ascent): - Gross liftoff weight, W_0 = 55,000 lbf - Propellant weight burned to cutoff, W_p = 40,000 lbf - Effective specific impulse, I_{sp} = 235 s (treated as constant — real engines vary I_{sp} with altitude, but a single effective value is the period-standard simplification for a first-pass performance estimate; see 1.2 for the non-constant version) - Burn time, t_b = 143 s - Gravity + drag loss allowance, Δv_losses = 3540 ft/s (derived in 1.1 from a first-principles gravity-turn/drag integration — treat it as a given here, the way a real FDO would receive it from the trajectory design group, having not run that study themselves)

Ballistic arc: - Burnout altitude, h_bo = 100,000 ft (fixed by the guidance program, independent of flight path angle — this matches the altitude 1.1's gravity-turn study actually converges to, which is not a coincidence: the pitch program in 1.1 was designed to hit it) - Standard gravity, g = 32.174 ft/s^2 (treated as constant with altitude — valid at these altitudes to within a few percent, and consistent with the flat, non-rotating Earth model used throughout this problem) - Assume impact at sea level (flat, non-rotating Earth — no curvature correction at this range; that correction is flagged as a natural extension for a later, longer-range stage)

Target: impact range = 210.0 nautical miles from the pad (1 nm = 6076.12 ft)

Authentic method + reference material

Powered ascent — Tsiolkovsky rocket equation (Tsiolkovsky, 1903; in active use by every rocket engineer of this era):

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Δv_ideal = c · ln(W₀/W₁)          c = Iₛₚ · g₀
V_bo = Δv_ideal − Δv_losses
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where $W_1 = W_0 - W_p$ is burnout weight.

Ballistic arc — flat-Earth, uniform-gravity vacuum trajectory. This is the same mathematics behind WWII- and V2-era artillery/rocket range tables, which the Redstone program inherited directly through von Braun's team. It is *not* the full spherical-Earth treatment (Bate, Mueller & White, Ch. 4, "Ballistic Missile Trajectories," which treats the free-flight arc as a conic with one focus at Earth's center) — that's intentionally saved for a later, longer-range stage, since Stage 1 is scoped to stay out of orbital mechanics. At suborbital ranges like this one, the flat-Earth form is an accurate and period-appropriate simplification, not a shortcut that sacrifices authenticity.

With burnout velocity V_bo at flight path angle γ (measured above local horizontal) and altitude h_bo:

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v_y0 = V_bo · sin(γ)        v_x0 = V_bo · cos(γ)

t_impact = [v_y0 + √(v_y0² + 2·g·h_bo)] / g

R = v_x0 · t_impact

h_apex = h_bo + v_y0² / (2g)
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The graphical range-table method: rather than solving R(γ) = 210.0 nm directly (it's not algebraically clean to invert), compute R for a spread of γ values, sketch R vs. γ by hand, and read the answer(s) off the curve. This is exactly how real range tables were built and used operationally — a chart, not an equation to invert on demand. Expect the curve to be non-monotonic near its peak, which means **two** valid flight path angles can produce the same range: a flatter, faster-arriving trajectory and a more lofted one. Same ambiguity in spirit as the SSA triangle case in 0.3 — don't be surprised by it, and don't discard one root as "wrong."

Worksheet

Part A — Burnout velocity

Quantity	Value
$W_1 = W_0 - W_p$	
$c = I_{sp} \cdot g_0$	
W_0/W_1	
$\ln(W_0/W_1)$	
$\Delta v\_ideal = c \cdot \ln(W_0/W_1)$	
$V\_bo = \Delta v\_ideal - \Delta v\_losses$	

Part B — Range table. Fill in for each γ. Carry v_y0^2 through in full (you'll reuse it for the apex altitude too — don't recompute).

γ	sin γ	cos γ	v_y0	v_x0	v_y0^2	t_impact (s)	R (ft)	R (nm)	h_apex (ft)
30°									
35°									

γ	sin γ	cos γ	v_y0	v_x0	v_{y0}^2	t_impact (s)	R (ft)	R (nm)	h_apex (ft)
40°									
45°									
50°									
55°									

Part C — Graphical solution. Plot R (nm, vertical axis) vs. γ (horizontal axis, 30°–55°) on the grid below using your six points from Part B. Draw a horizontal line at R = 210.0 nm and read off both crossings.

Low-angle solution: γ = __________° High-angle solution: γ = __________°

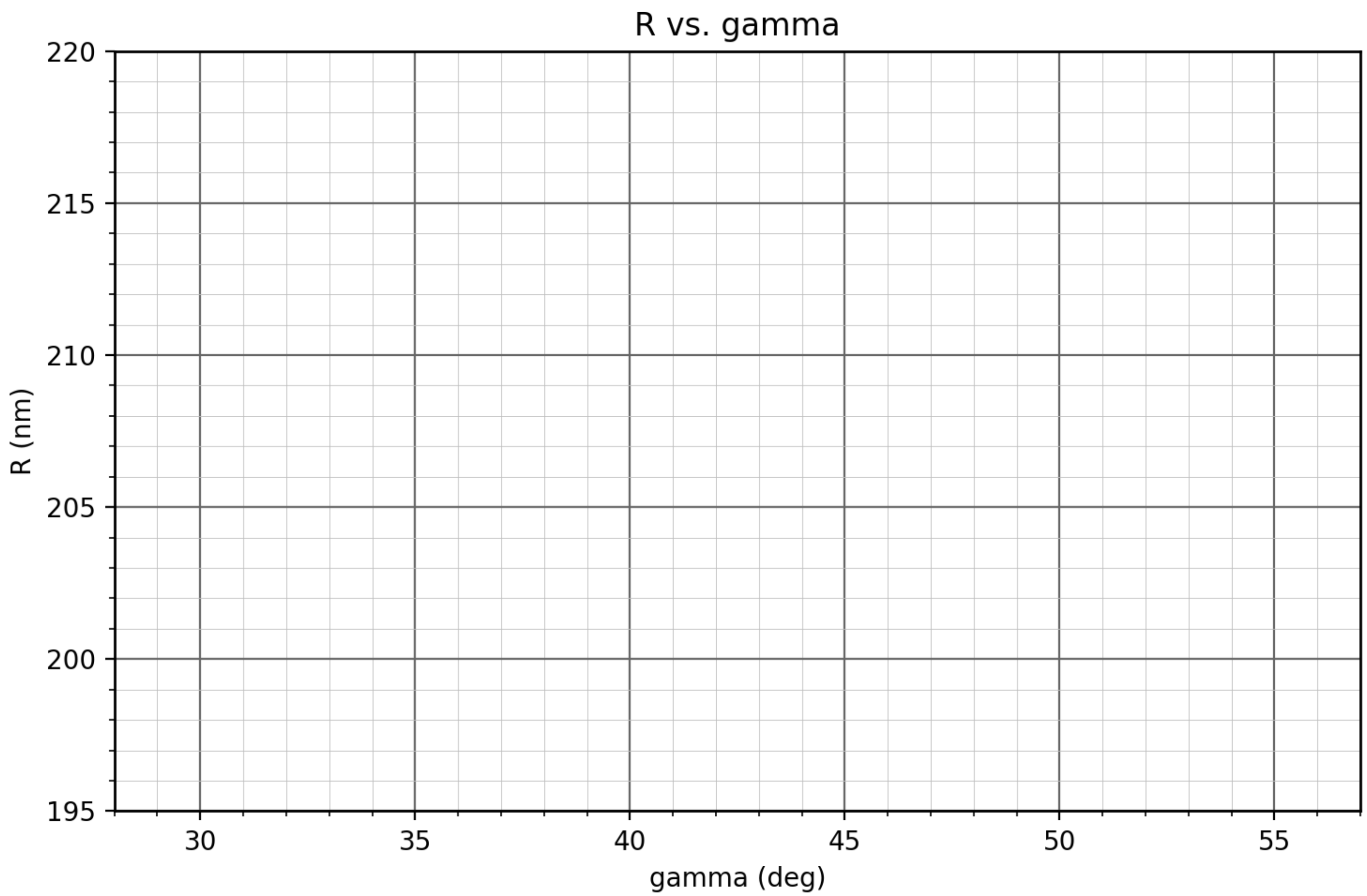

Figure 1: R vs. gamma